

Branch Integration and Bioplastic Class Extension

CNNIRPLASTICS — FTIR Polymer Classification

Jose Maria Contreras Prada

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Abstract

This report documents two changes made to the CNNIRPLASTICS repository: (1) the integration of the `origin/main` development line into the `add-cosine-baseline-eval` branch, unifying the preprocessing pipeline with the new random-forest analysis and reporting work; and (2) the extension of the classification target set with two bioplastics, PLA and PHA, at the data-builder and label-definition level. Both changes were committed and pushed to the remote.

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1 Branch Merge

1.1 Motivation

Prior to this work the branch `add-cosine-baseline-eval` and the `origin/main` branch had diverged: neither was a superset of the other. The feature branch carried the end-to-end pipeline and spectral preprocessing work (`run_pipeline.py`, `preprocess.py`, the cosine-similarity baseline, and the real-world lab samples), whereas `main` carried a reworked random-forest pipeline, single-spectrum test-time estimation, the feature-importance analysis, and an expanded requirements section in the README. Merging was required so that subsequent retraining used a single, unified code base.

1.2 Procedure

A safety branch `backup/pre-merge-add-cosine` was created before the merge. `origin/main` was then merged into `add-cosine-baseline-eval`. The merge produced content conflicts in three files, which were resolved manually as summarised in Table 1.

Table 1: Conflict resolution decisions.

File	Nature of conflict	Resolution
<code>models/cnn_model_draft.py</code>	Old monolithic <code>load_and_preprocess</code> (main) vs. refactored <code>create_model / train_model / test_model</code> (branch)	Kept branch: the pipeline depends on the refactored API
<code>models/rf_model.py</code>	<code>preprocess</code> -module routing (branch) vs. inline <code>min-max</code> plus feature-importance saving (main)	Kept main: preserves the new feature-importance analysis; its <code>min-max</code> is equivalent to <code>normalize="minmax"</code>
<code>data/format_data.py</code>	Four cosmetic print-string differences (unit notation, whitespace)	Kept branch: strings are already pure ASCII and stdout is forced to UTF-8

1.3 Verification

After resolution, the absence of conflict markers was confirmed, all core Python modules compiled, and the public APIs imported cleanly: the dataset loader, the preprocessing entry points (including SNV and AsLS/arPLS), and the refactored CNN functions. The six commodity-plastic classes (HDPE, LDPE, PP, PS, PVC, PET) were unchanged by the merge.

1.4 Outcome

The merge was recorded as commit `6cd97ea` and pushed to the remote. Files from both development lines are present in the unified tree: the pipeline and preprocessing work from the feature branch, and the random-forest, single-test, and requirements work from `main`. The pre-existing SNV and AsLS/arPLS implementations in `preprocess.py` were not modified and remain wired into the pipeline configuration.

2 Bioplastic Class Extension (PLA and PHA)

2.1 Background

The OpenSpecy spectral data committed to the repository contains only the six commodity plastics. A search of the 8390 metadata identity records returned zero matches for PLA- or PHA-related terms. This is by design: the dataset builder script downloads the raw OpenSpecy library, filters it down to the six target classes, and writes only the matched spectra to disk. The two bioplastics were therefore absent from every committed artefact.

2.2 Keyword Groups Added to the Dataset Builder

Two new keyword groups were added to the `target_polymers` list in `data/data_processing/openspecy_dataset.py`. Each polymer is matched against the lowercased `spectrum_identity` field using word-boundary regular expressions, so the new groups follow the same matching convention as the existing six.

```
PLA = c(
  "pla",
```

```

    "polylactic acid",
    "poly\\(lactic acid\\)",
    "polylactide",
    "lactic acid",
    "poly-l-lactic acid",
    "plla"
),

PHA = c(
    "pha",
    "phb",
    "phbv",
    "polyhydroxyalkanoate",
    "polyhydroxyalkanoates",
    "polyhydroxybutyrate",
    "poly\\(hydroxybutyrate\\)",
    "poly-3-hydroxybutyrate",
    "polyhydroxybutyrate-co-valerate"
)

```

The PHA group intentionally aggregates the PHB and PHBV variants, which are chemically members of the polyhydroxyalkanoate family, into a single class.

2.3 Label Definition Update

The two classes were registered in the canonical label list in `data/format_data.py`. Because the number of output classes is derived from the length of this list throughout the code base, this single edit resizes both the CNN softmax layer and the random-forest output space automatically.

```
POLYMER_CLASSES = ["HDPE", "LDPE", "PP", "PS", "PVC", "PET", "PLA", "PHA"]
```

This yields eight classes, with integer indices `PLA = 6` and `PHA = 7`.

2.4 Current Status and Caveat

The change is code-level scaffolding only. The two new classes remain *empty* until the OpenSpecy dataset is regenerated by re-running the builder script, which requires an R environment and a re-download of the raw OpenSpecy library. Until then no training spectrum carries a PLA or PHA label. The pipeline remains runnable in this state—the two classes simply contribute no samples— but a retrain should not be relied upon for bioplastic discrimination until the data is regenerated. In addition, while PLA is expected to be well represented in the OpenSpecy library, PHA/PHB/PHBV may be sparse; the class counts printed by the builder should be inspected before the class is trusted, and a second data source may be required.

2.5 Outcome

The keyword groups and the label update were recorded as commit `c20b143` and pushed to the remote.

3 Summary of Commits

4 Next Steps

1. Regenerate the OpenSpecy dataset by running the extended builder script in an R environment; inspect the per-class spectrum counts.

Table 2: Commits produced by this work, on branch `add-cosine-baseline-eval`.

Hash	Type	Description
6cd97ea	Merge	Integrate <code>origin/main</code> into the feature branch; resolve three conflicts
c20b143	Feature	Add PLA and PHA keyword groups and label definitions (scaffolding)

2. Clear the cached processed array (`output/_cache_processed.npz`) so the new eight-class data is reprocessed from scratch.
3. Retrain the CNN and random-forest models on the unified pipeline and evaluate per-class F1, with attention to the two new bioplastic classes.
4. If PHA coverage is insufficient, identify a supplementary FTIR source for that class.